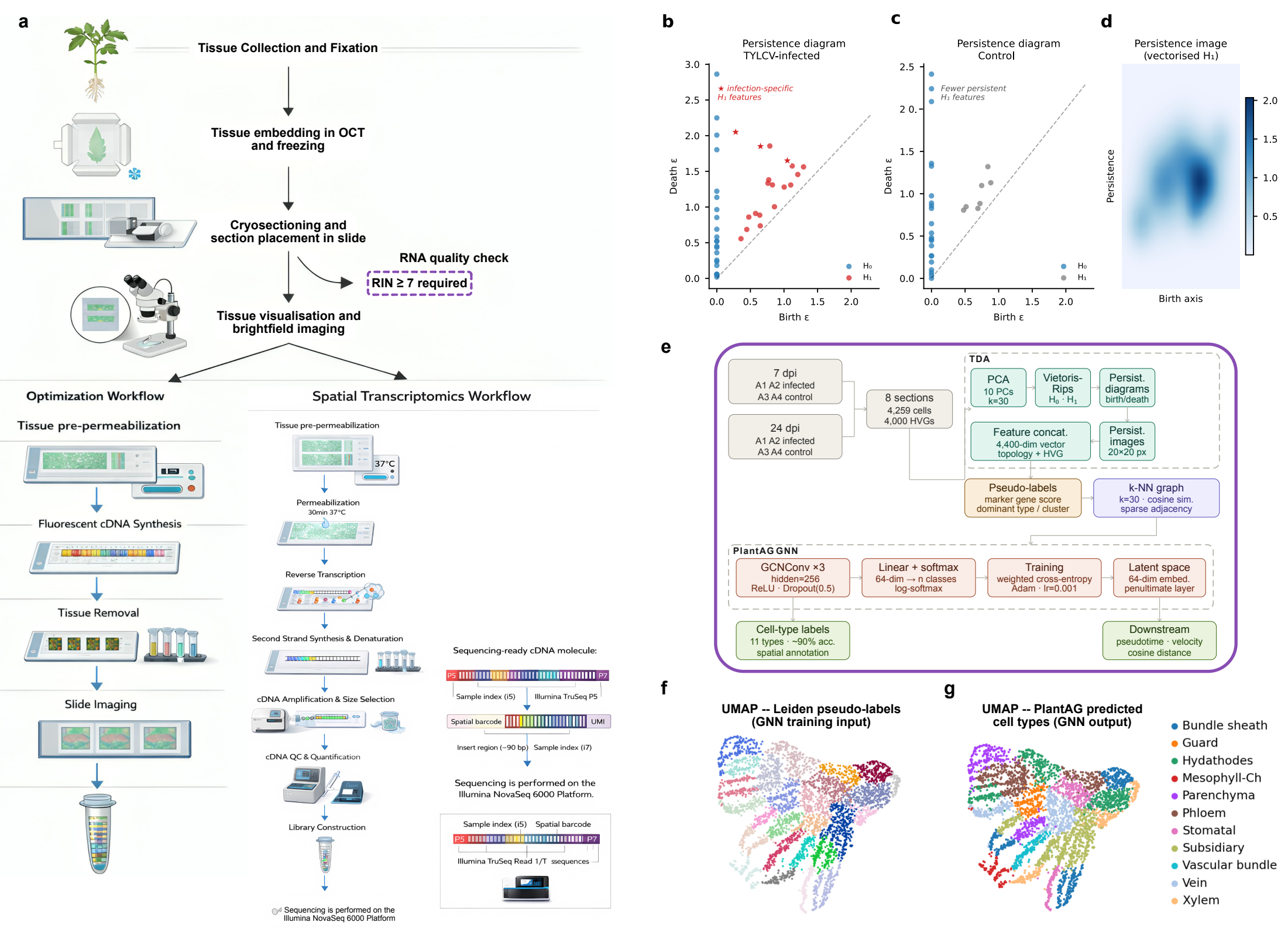

a
Tissue Collection and Fixation
Tissue embedding in OCT and freezing
Cryosectioning and section placement in slide
RNA quality check
RIN ≥ 7 required
Tissue visualisation and brightfield imaging
Optimization Workflow
Tissue pre-permeabilization
Fluorescent cDNA Synthesis
Tissue Removal
Slide Imaging
Spatial Transcriptomics Workflow
Tissue pre-permeabilization
37°C
Permeabilization
30min 37°C
Reverse Transcription
Second Strand Synthesis & Denaturation
cDNA Amplification & Size Selection
cDNA QC & Quantification
Library Construction
Sequencing is performed on the Illumina NovaSeq 6000 Platform
Sequencing-ready cDNA molecule:
P5
P7
Sample index (i5)
Illumina TruSeq P5
Spatial barcode
UMI
Insert region (~90 bp)
Sample index (i7)
Sequencing is performed on the Illumina NovaSeq 6000 Platform.
Sample index (i5)
Spatial barcode
Illumina TruSeq Read 1/T ssequences
b
Persistence diagram
TYLCV-infected
★ infection-specific H₁ features
Death ε
Birth ε
H₀
H₁
c
Persistence diagram
Control
Fewer persistent H₁ features
Death ε
Birth ε
H₀
H₁
d
Persistence image
(vectorised H₁)
Persistence
Birth axis
e
7 dpi
A1 A2 infected
A3 A4 control
24 dpi
A1 A2 infected
A3 A4 control
8 sections
4,259 cells
4,000 HVGs
TDA
PCA
10 PCs
k=30
Vietoris-Rips
H₀ · H₁
Persist. diagrams
birth/death
Feature concat.
4,400-dim vector
topology + HVG
Persist. images
20×20 px
Pseudo-labels
marker gene score
dominant type / cluster
k-NN graph
k=30 · cosine sim.
sparse adjacency
PlantAG GNN
GCNConv ×3
hidden=256
ReLU · Dropout(0.5)
Linear + softmax
64-dim → n classes
log-softmax
Training
weighted cross-entropy
Adam · lr=0.001
Latent space
64-dim embed.
penultimate layer
Cell-type labels
11 types · ~90% acc.
spatial annotation
Downstream
pseudotime · velocity
cosine distance
f
UMAP -- Leiden pseudo-labels
(GNN training input)
g
UMAP -- PlantAG predicted
cell types (GNN output)
Bundle sheath
Guard
Hydathodes
Mesophyll-Ch
Parenchyma
Phloem
Stomatal
Subsidiary
Vascular bundle
Vein
Xylem